

University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Term Final Examination, Spring 2025

Course Title: Digital Logic Design

Course Code: CSE 0611215/CSE215/CSE201

Marks: 50

Time: 3(three)hours

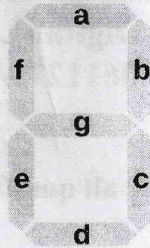
(Answer all questions)

- | Q. No. | | Marks |
|--------|--|-------|
| 1. a) | Simplify the following boolean function F with the don't-care conditions d, and then express the simplified function in <u>sum-of-minterms</u> and <u>product of maxterms</u> . | 4 |
| | $F(A, B, C, D) = \Sigma(5, 6, 7, 12, 14, 15)$ $d(A, B, C, D) = \Sigma(13, 9, 11, 10)$ | |
| b) | A display operates using Excess-3 code instead of standard BCD to simplify certain internal processing. Now, you are required to implement a code converter that converts a decimal digit input (in BCD format) to its Excess-3 code equivalent. | 6 |
| 2. a) | Suppose you have two numbers, $A = (9)_{10}$ and $B = (7)_{10}$. Convert them into 4-bit binary representations. Using a 4-bit parallel adder, perform bitwise addition with carry propagation. Since the result may require an extra bit for carry-out, illustrate a full adder circuit for each bit, showing inputs, carry propagation, and final sum output. | 4 |
| b) | Discuss about the combinational circuit named half subtractor with a proper truth table and a circuit diagram. | 2 |
| c) | Construct a 4-to-16-line decoder with five 2-to-4-line decoders with an enable input for implementing the following function. | 4 |
| | $F(A, B, C, D) = \Sigma(0, 2, 4, 6, 11, 13).$ | |
| 3. a) | Evaluate the following boolean function using a 4x1 MUX. | 5 |
| | $F(A, B, C, D) = \Sigma(1, 3, 4, 7, 11, 12, 13, 14, 15)$ | |

b)

A seven-segment display is used to represent the letters H, E, L, and P.

5

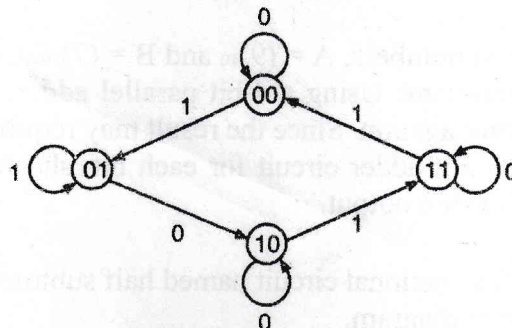


Determine the logical expression for segment 'a' using a truth table and K-map simplification. The BCD value for H, E, L, P will be 0,1,2,3

4. a) Describe the step-by-step process of converting a Full adder into a Full subtractor, highlighting the necessary modifications in logic gates and input configurations. 5

- b) Design a D flip-flop by using JK flip-flop. 5

5. a) Construct a sequential circuit from the following state diagram (use JK F/F) 10



J K